

Fourier Series and Partial Differential Equations

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Unit 18: From Fourier Series to Separation of Variables and the Classical PDEs

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Unit 18 at a Glance: Fourier Series as the Key to Partial Differential Equations

A **Fourier series** expresses a periodic function as a sum of sines and cosines, and because the trigonometric system is *orthogonal*, every coefficient is one independent integral — the **Euler–Fourier formulas**. Symmetry simplifies the work: **even** functions need only cosines, **odd** functions only sines, and a function on a half-interval can be expanded as either via a half-range extension. This is the toolkit that cracks **partial differential equations**: **separation of variables** $u = X(x)T(t)$ splits a PDE into ODEs linked by a separation constant, the boundary conditions reproduce the eigenvalue BVP of Unit 17, and the initial profile is matched by a Fourier series. The **Laplacian** $\nabla^2 u$ governs the three classical equations — heat (diffusion), wave (vibration), and Laplace (steady state).

- **Fourier intuition** — any reasonable periodic signal is a superposition of pure frequencies; the Fourier transform is the continuous-spectrum limit of the Fourier series.
- **Computing the series** — $a_n = \frac{1}{L} \int_{-L}^L f \cos \frac{n\pi x}{L} dx$, $b_n = \frac{1}{L} \int_{-L}^L f \sin \frac{n\pi x}{L} dx$, derived from trig orthogonality.
- **Symmetry and half-range** — even $\Rightarrow$ cosine series, odd $\Rightarrow$ sine series; extend a function on $0 \leq x \leq L$ evenly or oddly to get either.
- **Separation of variables** — $u(x, t) = X(x)T(t)$ turns a linear homogeneous PDE into two ODEs; the spatial ODE is the Unit 17 eigenvalue problem.
- **The Laplacian and the classical PDEs** — $\nabla^2 u = u_{xx} + u_{yy}$ drives heat $u_t = \alpha^2 u_{xx}$, wave $u_{tt} = c^2 u_{xx}$, and Laplace $\nabla^2 u = 0$.

18.1 Fourier Series and Transform Intuition

A periodic function as a superposition of pure-frequency sines and cosines; the time-domain versus frequency-domain picture; harmonics and the fundamental frequency; the Fourier transform as the

continuous-spectrum limit of the Fourier series.

The Fourier Series — Superposition of Harmonics

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right), \quad \text{period } 2L.$$

Harmonics and the Fundamental Frequency

the n th harmonic has frequency $\frac{n}{2L}$;
 $n = 1$ is the *fundamental*, $n \geq 2$ the *overtones*.

The Fourier Transform — Continuous Spectrum

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi x} dx;$$

the $L \rightarrow \infty$ limit turns the discrete harmonics $n/2L$ into a continuum ξ .

Method — Read a Function in the Frequency Domain

1. Treat the signal $f(x)$ as living in the *time* (or space) domain — its value at each point.
2. Ask which pure frequencies $\cos(n\pi x/L)$, $\sin(n\pi x/L)$ it is built from, and with what amplitudes a_n, b_n .
3. Plot the amplitudes against frequency: this *frequency-domain* spectrum is an equivalent description of the same signal.
4. For a non-periodic signal, let the period $2L \rightarrow \infty$; the discrete spectrum fills in to a continuous one and the sum becomes the Fourier transform integral.

Worked Example — Spectrum of a Two-Tone Signal

Identify the Fourier content of $f(x) = 3 + 5 \cos(\pi x) - 2 \sin(3\pi x)$ on the interval of period 2 (so $L = 1$).

Match the template: $f(x) = \frac{a_0}{2} + \sum_n (a_n \cos n\pi x + b_n \sin n\pi x)$ with $L = 1$.

Constant term: $\frac{a_0}{2} = 3 \Rightarrow a_0 = 6$ (the average / DC level).

Cosine amplitude: $a_1 = 5$ (a fundamental cosine at frequency $\frac{1}{2}$).

Sine amplitude: $b_3 = -2$ (a third-harmonic sine); all other $a_n, b_n = 0$.

Spectrum: only three lines — DC, $n = 1$ cosine, $n = 3$ sine — because f is already a finite trig sum.

Key Takeaway

A Fourier series says any reasonable periodic function is a **superposition of pure frequencies** — a chord of sines and cosines. The same signal then has two equivalent portraits: the familiar **time-domain** graph and the **frequency-domain** spectrum of amplitudes. Letting the period grow without bound turns the discrete spectrum into a continuous one, and the series becomes the Fourier transform.

Common Pitfall

The constant term is $a_0/2$, **not** a_0 — the factor of $\frac{1}{2}$ is what makes the single Euler–Fourier formula $a_n = \frac{1}{L} \int_{-L}^L f \cos \frac{n\pi x}{L} dx$ work for $n = 0$ as well. Also, a Fourier *series* needs a **periodic** (or finite-interval) function; a non-periodic signal on the whole line requires the Fourier *transform*, not a series.

18.2 Computing Fourier Series

The Euler–Fourier coefficient formulas on $-L \leq x \leq L$; orthogonality of the trigonometric system as their engine; the constant (DC) term $a_0/2$; convergence at jumps to the midpoint of the one-sided limits (Dirichlet conditions).

The Euler–Fourier Coefficient Formulas

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx \quad (n \geq 0), \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx \quad (n \geq 1).$$

Orthogonality of the Trigonometric System

$$\int_{-L}^L \cos \frac{n\pi x}{L} \cos \frac{m\pi x}{L} dx = L \delta_{nm}, \quad \int_{-L}^L \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx = L \delta_{nm}, \quad \int_{-L}^L \cos \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx = 0.$$

Convergence at a Jump (Dirichlet)

At a jump x_0 , the series converges to $\frac{1}{2} [f(x_0^-) + f(x_0^+)]$, the midpoint of the one-sided limits.

Method — Compute a Fourier Series

1. Identify the half-period L so the function repeats with period $2L$, and write the target $f = \frac{a_0}{2} + \sum_n (a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L})$.
2. Compute $a_0 = \frac{1}{L} \int_{-L}^L f dx$ — twice the average value of f over one period.
3. Compute a_n and b_n from the Euler–Fourier integrals, using integration by parts where f is polynomial times trig.
4. Exploit symmetry to skip half the work: if f is even all $b_n = 0$; if f is odd all $a_n = 0$ (and $a_0 = 0$).
5. State the series and note its value at any jump is the midpoint $\frac{1}{2}[f(x_0^-) + f(x_0^+)]$.

Worked Example — Fourier Series of $f(x) = x$ on $-\pi \leq x \leq \pi$

Find the Fourier series of the 2π -periodic extension of $f(x) = x$ on $-\pi \leq x \leq \pi$ (so $L = \pi$), computing every coefficient in full.

Symmetry: $f(x) = x$ is odd $\Rightarrow a_0 = 0$ and $a_n = 0$ for all n ; only the b_n survive.

Set up b_n : $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(nx) dx = \frac{2}{\pi} \int_0^{\pi} x \sin(nx) dx$ (even integrand).

By parts: $u = x, dv = \sin(nx) dx \Rightarrow du = dx, v = -\frac{1}{n} \cos(nx)$.

$$\int_0^{\pi} x \sin(nx) dx = \left[-\frac{x}{n} \cos(nx) \right]_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos(nx) dx = -\frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} [\sin(nx)]_0^{\pi}.$$

Evaluate: $\cos(n\pi) = (-1)^n, \sin(n\pi) = 0 \Rightarrow \int_0^{\pi} x \sin(nx) dx = -\frac{\pi}{n} (-1)^n = (-1)^{n+1} \frac{\pi}{n}$.

Coefficient: $b_n = \frac{2}{\pi} \cdot (-1)^{n+1} \frac{\pi}{n} = (-1)^{n+1} \frac{2}{n}$.

Series: $x = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2}{n} \sin(nx) = 2 \left(\sin x - \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x - \cdots \right)$, converging to 0 at $x = \pm\pi$.

Key Takeaway

The Euler–Fourier formulas are pure **orthogonality**: integrating f against a single basis wave annihilates every other term and leaves one coefficient. The constant $a_0/2$ is the function's **average**; the a_n, b_n are how strongly each harmonic is present. Symmetry is the great labor-saver — recognizing f as even or odd erases half the integrals before you compute anything.

Common Pitfall

Watch the normalizations: the coefficients use $\frac{1}{L}$ (over the full period $2L$), while the half-range formulas of the next module use $\frac{2}{L}$ (over $0 \leq x \leq L$). **Mixing the two** is the most common Fourier error. And the series equals f only where f is continuous — at a **jump** it converges to the midpoint $\frac{1}{2}[f(x_0^-) + f(x_0^+)]$, not to either one-sided value.

18.3 Fourier Series for Differential Equations

Even and odd functions and their product rules; the pure cosine series (even) and pure sine series (odd); half-range expansions via even or odd extension of a function given only on $0 \leq x \leq L$; choosing the expansion that matches the boundary conditions of a PDE.

Even/Odd Symmetry Collapses the Series

$$f \text{ even} \Rightarrow b_n = 0 \text{ (pure cosine series);} \quad f \text{ odd} \Rightarrow a_n = 0 \text{ (pure sine series, and } a_0 = 0).$$

Half-Range Cosine Series (Even Extension)

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L}, \quad a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx \quad (n \geq 0).$$

Half-Range Sine Series (Odd Extension)

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}, \quad b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx.$$

Method — Build a Half-Range Expansion

1. You are given f only on $0 \leq x \leq L$ and must represent it by sines *or* cosines alone.
2. For a *sine* series, extend f as an *odd* function to $-L \leq x \leq 0$; for a *cosine* series, extend it as an *even* function.
3. Apply the half-range formula with the $\frac{2}{L}$ normalization (the $-L$ to 0 contribution doubles the 0 to L integral).
4. Let the boundary conditions of the target PDE pick the series: Dirichlet ends ($u = 0$) call for the *sine* series, Neumann ends ($u_x = 0$) for the *cosine* series.
5. Write the expansion, remembering the odd extension forces $f(0) = f(L) = 0$ in the limit, matching homogeneous Dirichlet data.

Worked Example — Half-Range Sine Series of $f(x) = 1$ on $0 \leq x \leq L$

Expand the constant $f(x) = 1$ on $0 \leq x \leq L$ in a half-range *sine* series (odd extension), the form needed for a Dirichlet BVP.

$$\text{Formula: } b_n = \frac{2}{L} \int_0^L 1 \cdot \sin \frac{n\pi x}{L} dx.$$

$$\text{Integrate: } \int_0^L \sin \frac{n\pi x}{L} dx = \left[-\frac{L}{n\pi} \cos \frac{n\pi x}{L} \right]_0^L = -\frac{L}{n\pi} (\cos n\pi - 1).$$

$$\text{Simplify: } \cos n\pi = (-1)^n \Rightarrow \int_0^L \sin \frac{n\pi x}{L} dx = \frac{L}{n\pi} (1 - (-1)^n).$$

$$\text{Coefficient: } b_n = \frac{2}{L} \cdot \frac{L}{n\pi} (1 - (-1)^n) = \frac{2}{n\pi} (1 - (-1)^n) = \begin{cases} \frac{4}{n\pi}, & n \text{ odd,} \\ 0, & n \text{ even.} \end{cases}$$

$$\text{Series: } 1 = \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{1}{2k+1} \sin \frac{(2k+1)\pi x}{L} = \frac{4}{\pi} \left(\sin \frac{\pi x}{L} + \frac{1}{3} \sin \frac{3\pi x}{L} + \cdots \right), \quad 0 < x < L.$$

Key Takeaway

Symmetry is a choice you can **impose**. A function living only on $0 \leq x \leq L$ has no symmetry of its own, but you may extend it **evenly** to get a cosine series or **oddly** to get a sine series. Which you pick is dictated by the PDE: **Dirichlet** boundary data wants the **sine** series (it vanishes at the ends), **Neumann** data wants the **cosine** series (its slope vanishes).

Common Pitfall

The half-range formulas carry $\frac{2}{L}$, **not** $\frac{1}{L}$ — the odd or even extension doubles the integral over $0 \leq x \leq L$. Do not also integrate over $-L$ to L ; that double-counts. And the sine series of a function with $f(0) \neq 0$ (like $f = 1$ above) **converges to 0 at the endpoints** because the odd extension forces a jump there — expected, not an error.

18.4 Separable Partial Differential Equations

The separation-of-variables ansatz $u(x, t) = X(x)T(t)$; splitting a linear homogeneous PDE into two ODEs joined by a separation constant; the role of homogeneous boundary conditions in producing the spatial eigenvalue BVP; the classical PDEs and their boundary/initial data.

The Separation Ansatz and Constant

$$u(x, t) = X(x)T(t);$$

$$\frac{1}{\alpha^2} \frac{T'}{T} = \frac{X''}{X} = -\lambda \quad (\text{constant: each side depends on a different variable}).$$

The Two Resulting ODEs (Heat Equation)

$$X'' + \lambda X = 0 \text{ (spatial, an eigenvalue BVP);} \quad T' + \alpha^2 \lambda T = 0 \text{ (temporal, first order, } T_n = e^{-\alpha^2 \lambda_n t}).$$

Superposition of the Product Solutions

$$u(x, t) = \sum_{n=1}^{\infty} b_n X_n(x) T_n(t);$$

the b_n are fixed by matching the initial condition with a Fourier series.

Method — Solve a PDE by Separation of Variables

1. Assume a product solution $u(x, t) = X(x)T(t)$ and substitute into the PDE.
2. Divide through so all x -dependence is on one side and all t -dependence on the other; set both equal to the separation constant $-\lambda$.
3. Solve the *spatial* ODE with the homogeneous boundary conditions first — this is the Unit 17 eigenvalue BVP, yielding eigenvalues λ_n and eigenfunctions X_n .
4. Solve the *temporal* ODE for each λ_n to get $T_n(t)$, then form the superposition $u = \sum_n b_n X_n T_n$.
5. Impose the initial condition $u(x, 0) = f(x)$ and read off the b_n as the Fourier (half-range) coefficients of f .

Worked Example — Separating the Heat Equation

Separate $u_t = \alpha^2 u_{xx}$ on $0 \leq x \leq L$ with $u(0, t) = u(L, t) = 0$, and identify the two ODEs and the spatial eigenpairs.

$$\text{Substitute } u = X(x)T(t) : XT' = \alpha^2 X''T.$$

$$\text{Divide by } \alpha^2 XT : \frac{T'}{\alpha^2 T} = \frac{X''}{X}.$$

Separation constant: LHS depends only on t , RHS only on $x \Rightarrow$ both $= -\lambda$.

Spatial BVP: $X'' + \lambda X = 0$, $X(0) = X(L) = 0 \Rightarrow \lambda_n = \left(\frac{n\pi}{L}\right)^2$, $X_n = \sin \frac{n\pi x}{L}$ (Unit 17).

$$\text{Temporal ODE: } T' + \alpha^2 \lambda_n T = 0 \Rightarrow T_n(t) = e^{-\alpha^2 (n\pi/L)^2 t}.$$

Product modes: $u_n(x, t) = \sin \frac{n\pi x}{L} e^{-\alpha^2 (n\pi/L)^2 t}$, ready to superpose.

Key Takeaway

Separation of variables is the master move for linear PDEs: assuming $u = X(x)T(t)$ **decouples** the equation into two ODEs tied by a single separation constant. The homogeneous boundary conditions turn the spatial ODE into the **eigenvalue BVP of Unit 17**, so its eigenfunctions become the building blocks; **superposition** assembles them and the **initial condition** fixes the weights through a Fourier series.

Common Pitfall

Separation of variables needs the PDE to be **linear and homogeneous with homogeneous boundary conditions** — a nonzero boundary value or a forcing term breaks the clean split (handle those by first subtracting a steady-state/particular solution). Also, always solve the **spatial** eigenvalue problem before the temporal one: the boundary conditions live in x , and they are what quantize λ .

18.5 The Laplacian

The Laplacian $\nabla^2 u = \nabla \cdot \nabla u = u_{xx} + u_{yy}$ as the divergence of the gradient; its meaning as how a value departs from its local average; Laplace's equation $\nabla^2 u = 0$ and harmonic functions; the steady-state limit of diffusion; polar/spherical forms.

The Laplacian — Divergence of the Gradient

$$\nabla^2 u = \nabla \cdot (\nabla u) = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \quad (\text{Cartesian, here through 3D}).$$

Laplace's Equation and Harmonic Functions

$$\nabla^2 u = 0 \quad \Longleftrightarrow \quad u \text{ is harmonic};$$

value at a point = average over any surrounding circle/sphere (mean value).

The Laplacian in Polar Coordinates

$$\nabla^2 u = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} \quad (\text{the natural form for disks and annuli}).$$

Method — Test and Interpret the Laplacian

1. Compute the second partial in each spatial variable and add them to form $\nabla^2 u$.
2. Read the sign as curvature versus the neighborhood: $\nabla^2 u > 0$ means the value sits *below* its local average (a dip), $\nabla^2 u < 0$ means *above* (a bump).
3. If $\nabla^2 u = 0$ everywhere, the function is harmonic — it equals its local average and has no interior maxima or minima (maximum principle).
4. For diffusion, recognize $\nabla^2 u$ as the spatial driver: heat flows so as to erase departures from the average, and the steady state ($u_t = 0$) is exactly Laplace's equation.

Worked Example — Verify a Harmonic Function

Show that $u(x, y) = x^2 - y^2$ is harmonic, and contrast it with $v(x, y) = x^2 + y^2$.

Gradient pieces: $u_x = 2x$, $u_{xx} = 2$; $u_y = -2y$, $u_{yy} = -2$.

Laplacian of u : $\nabla^2 u = u_{xx} + u_{yy} = 2 + (-2) = 0 \Rightarrow u$ is harmonic.

Now v : $v_{xx} = 2$, $v_{yy} = 2 \Rightarrow \nabla^2 v = 4 \neq 0 \Rightarrow v$ is *not* harmonic.

Interpretation: $\nabla^2 v = 4 > 0 \Rightarrow v$ sits below its local average — a bowl with an interior minimum.

Contrast: $u = x^2 - y^2$ is a saddle; its curvatures cancel, so no interior extremum (maximum principle).

Key Takeaway

The Laplacian $\nabla^2 u$ is the **divergence of the gradient** — a scalar measuring how a point's value departs from the **average of its neighbors**. Where it is positive the point is a relative dip, where negative a relative bump, and where it is **zero the function is harmonic** and equals its local average. This single operator is the spatial engine of all three classical PDEs.

Common Pitfall

The Laplacian is a *sum* of unmixed second partials, $u_{xx} + u_{yy}$ — **do not include the mixed partial** u_{xy} , and do not confuse $\nabla^2 u$ (a scalar) with the gradient ∇u (a vector) or the Hessian (a matrix). In curved coordinates you **cannot** just sum naive second derivatives: polar form carries the extra $\frac{1}{r}u_r$ and $\frac{1}{r^2}u_{\theta\theta}$ terms.

18.6 The Heat and Wave Equations

The heat equation $u_t = \alpha^2 u_{xx}$ (parabolic diffusion) solved by separation plus a Fourier sine series; the wave equation $u_{tt} = c^2 u_{xx}$ (hyperbolic vibration) with standing-mode and d'Alembert traveling-wave solutions; Duhamel's principle for a forced wave; the classification of the three classical equations.

The Three Classical Equations

Heat (parabolic): $u_t = \alpha^2 u_{xx}$;

Wave (hyperbolic): $u_{tt} = c^2 u_{xx}$;

Laplace (elliptic): $u_{xx} + u_{yy} = 0$.

Heat Equation — Fourier Solution

$$u(x, t) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} e^{-\alpha^2 (n\pi/L)^2 t}, \quad b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx.$$

Wave Equation — d'Alembert's Solution

$$u(x, t) = \frac{1}{2} [f(x - ct) + f(x + ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds \quad (\text{initial shape } f, \text{ initial velocity } g).$$

Method — Solve the 1D Heat Equation on a Rod

1. Separate $u = X(x)T(t)$ and solve the spatial BVP $X'' + \lambda X = 0$, $X(0) = X(L) = 0$ to get $\lambda_n = (n\pi/L)^2$, $X_n = \sin \frac{n\pi x}{L}$.
2. Solve $T' + \alpha^2 \lambda_n T = 0$ for $T_n = e^{-\alpha^2 (n\pi/L)^2 t}$, so each mode decays faster the higher its frequency.
3. Superpose: $u(x, t) = \sum_n b_n \sin \frac{n\pi x}{L} e^{-\alpha^2 (n\pi/L)^2 t}$.
4. Set $t = 0$: $u(x, 0) = \sum_n b_n \sin \frac{n\pi x}{L} = f(x)$, so the b_n are the half-range *sine* coefficients of the initial temperature f .
5. Evaluate $b_n = \frac{2}{L} \int_0^L f \sin \frac{n\pi x}{L} dx$ and write the full series solution.

Worked Example — Heat on a Rod with Initial Temperature $f(x) = 100$

A rod $0 \leq x \leq L$ has ends held at 0° ($u(0, t) = u(L, t) = 0$) and uniform initial temperature $u(x, 0) = 100$. Solve $u_t = \alpha^2 u_{xx}$ completely, computing the coefficients in full.

Separate: $u = XT \Rightarrow X'' + \lambda X = 0$, $X(0) = X(L) = 0$, and $T' + \alpha^2 \lambda T = 0$.

Spatial eigenpairs: $\lambda_n = \left(\frac{n\pi}{L}\right)^2$, $X_n = \sin \frac{n\pi x}{L}$, $n = 1, 2, 3, \dots$

Temporal factor: $T_n(t) = e^{-\alpha^2 (n\pi/L)^2 t}$.

Superposition: $u(x, t) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} e^{-\alpha^2 (n\pi/L)^2 t}$.

Match $t = 0$: $\sum_n b_n \sin \frac{n\pi x}{L} = 100 \Rightarrow b_n = \frac{2}{L} \int_0^L 100 \sin \frac{n\pi x}{L} dx$.

Integrate: $\int_0^L \sin \frac{n\pi x}{L} dx = \frac{L}{n\pi} (1 - (-1)^n) \Rightarrow b_n = \frac{200}{n\pi} (1 - (-1)^n)$.

Odd-only: $b_n = \frac{400}{n\pi}$ (n odd), 0 (n even).

Solution: $u(x, t) = \frac{400}{\pi} \sum_{k=0}^{\infty} \frac{1}{2k+1} \sin \frac{(2k+1)\pi x}{L} e^{-\alpha^2 ((2k+1)\pi/L)^2 t} \xrightarrow{t \rightarrow \infty} 0$.

Key Takeaway

The classical PDEs all yield to the same pipeline: **separate**, solve the **spatial eigenvalue BVP**, attach the time factor, **superpose**, and fit the initial data with a **Fourier series**. Their characters differ — the **heat** equation smooths and decays (high modes die fastest), the **wave** equation oscillates and propagates (d'Alembert's two traveling waves), and **Laplace** gives the steady state — but Fourier analysis is the common engine.

Common Pitfall

Match the time factor to the equation's order in t : the heat equation is *first* order so each mode carries a single decaying $e^{-\alpha^2 \lambda_n t}$, while the wave equation is *second* order so each mode needs **two** time functions $\cos(c\sqrt{\lambda_n} t)$ and $\sin(c\sqrt{\lambda_n} t)$ — and therefore *two* initial conditions (shape *and* velocity). Using a single exponential for the wave equation **loses half the solution**.

Practice Problems

Work the following ten problems in order. They climb from reading the harmonics of a signal and computing Euler–Fourier coefficients, through even/odd symmetry and half-range expansions, into separation of variables and the Laplacian, and up to a complete separation-of-variables solution of the heat equation with every coefficient integral expanded, plus the d’Alembert solution of the wave equation. Complete solutions for all problems follow.

1. **(Harmonics)** For $f(x) = 4 - 3\cos(2\pi x) + \sin(\pi x)$ with period 2 ($L = 1$), state a_0 , the nonzero a_n , and the nonzero b_n .
2. **(Euler–Fourier setup)** Write the Euler–Fourier formulas for a_n and b_n of a $2L$ -periodic function, and state the orthogonality relations that justify them.
3. **(Odd function series)** Find the Fourier series of the 2π -periodic extension of $f(x) = x$ on $-\pi \leq x \leq \pi$, computing b_n in full.
4. **(Symmetry shortcut)** Without integrating, explain why $f(x) = x^2$ on $-\pi \leq x \leq \pi$ has $b_n = 0$ for all n , and which series type it has.
5. **(Half-range sine)** Expand $f(x) = 1$ on $0 \leq x \leq L$ in a half-range sine series, giving b_n and the resulting series.
6. **(Separation of variables)** Separate $u_t = \alpha^2 u_{xx}$ via $u = X(x)T(t)$; state the two ODEs and, with $u(0, t) = u(L, t) = 0$, the spatial eigenpairs and time factor.
7. **(Laplacian / harmonic)** Determine whether $u(x, y) = x^2 - y^2$ and $v(x, y) = e^x \sin y$ are harmonic by computing ∇^2 of each.
8. **(Classification)** Classify $u_t = \alpha^2 u_{xx}$, $u_{tt} = c^2 u_{xx}$, and $u_{xx} + u_{yy} = 0$ as parabolic, hyperbolic, or elliptic, and name the phenomenon each models.
9. **(Heat equation capstone)** Solve $u_t = \alpha^2 u_{xx}$ on $0 \leq x \leq L$, $u(0, t) = u(L, t) = 0$, $u(x, 0) = 100$, computing the coefficients in full.
10. **(Wave equation)** Use d’Alembert’s formula to solve $u_{tt} = c^2 u_{xx}$ on the line with $u(x, 0) = f(x)$ and $u_t(x, 0) = 0$, and describe the motion.

Complete Solutions

Solution 1. Match $f = \frac{a_0}{2} + \sum_n (a_n \cos n\pi x + b_n \sin n\pi x)$ with $L = 1$. The constant $\frac{a_0}{2} = 4$, so

$$a_0 = 8; \quad a_2 = -3 \text{ (from } -3 \cos 2\pi x); \quad b_1 = 1 \text{ (from } \sin \pi x).$$

All other coefficients are 0.

Solution 2. For a $2L$ -periodic f ,

$$a_n = \frac{1}{L} \int_{-L}^L f \cos \frac{n\pi x}{L} dx, \quad b_n = \frac{1}{L} \int_{-L}^L f \sin \frac{n\pi x}{L} dx.$$

They follow from orthogonality: $\int_{-L}^L \cos \frac{n\pi x}{L} \cos \frac{m\pi x}{L} dx = L\delta_{nm}$, the same for sines, and $\int_{-L}^L \cos \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx = 0$ — integrating f against one basis wave isolates its coefficient.

Solution 3. $f(x) = x$ is odd, so $a_n = 0$ and only b_n survive. With $L = \pi$, integration by parts gives

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(nx) dx = \frac{2}{\pi} \left[-\frac{\pi}{n}(-1)^n \right] = (-1)^{n+1} \frac{2}{n}.$$

Hence $x = \sum_{n \geq 1} (-1)^{n+1} \frac{2}{n} \sin(nx) = 2(\sin x - \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x - \dots)$.

Solution 4. $f(x) = x^2$ is *even* ($f(-x) = f(x)$). The integrand $x^2 \sin \frac{n\pi x}{L}$ is then odd, so

$$b_n = \frac{1}{L} \int_{-L}^L x^2 \sin \frac{n\pi x}{L} dx = 0 \quad (\text{odd integrand over a symmetric interval}).$$

It therefore has a pure *cosine* series (only a_0 and a_n).

Solution 5. Half-range sine: $b_n = \frac{2}{L} \int_0^L \sin \frac{n\pi x}{L} dx$. Then

$$b_n = \frac{2}{L} \cdot \frac{L}{n\pi} (1 - (-1)^n) = \frac{2}{n\pi} (1 - (-1)^n) = \frac{4}{n\pi} \text{ (} n \text{ odd), } 0 \text{ (} n \text{ even)}.$$

So $1 = \frac{4}{\pi} \sum_{k \geq 0} \frac{1}{2k+1} \sin \frac{(2k+1)\pi x}{L}$ on $0 < x < L$.

Solution 6. Substituting $u = XT$ into $u_t = \alpha^2 u_{xx}$ and dividing by $\alpha^2 XT$ gives $\frac{T'}{\alpha^2 T} = \frac{X''}{X} = -\lambda$. Thus

$$X'' + \lambda X = 0, \quad X(0) = X(L) = 0 \Rightarrow \lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad X_n = \sin \frac{n\pi x}{L};$$

$$T' + \alpha^2 \lambda_n T = 0 \Rightarrow T_n = e^{-\alpha^2 (n\pi/L)^2 t}.$$

Solution 7. For $u = x^2 - y^2$: $u_{xx} = 2$, $u_{yy} = -2$, so

$$\nabla^2 u = 2 - 2 = 0 \text{ (harmonic)}.$$

For $v = e^x \sin y$: $v_{xx} = e^x \sin y$, $v_{yy} = -e^x \sin y$, so

$$\nabla^2 v = e^x \sin y - e^x \sin y = 0 \text{ (also harmonic)}.$$

Solution 8. By the second-order classification:

$$u_t = \alpha^2 u_{xx} : \text{ parabolic (diffusion/heat),}$$

$$u_{tt} = c^2 u_{xx} : \text{ hyperbolic (vibration/waves),}$$

$$u_{xx} + u_{yy} = 0 : \text{ elliptic (steady state).}$$

Solution 9. Separation gives $X_n = \sin \frac{n\pi x}{L}$, $T_n = e^{-\alpha^2(n\pi/L)^2 t}$, so $u = \sum_n b_n \sin \frac{n\pi x}{L} e^{-\alpha^2(n\pi/L)^2 t}$. Matching $u(x, 0) = 100$,

$$b_n = \frac{2}{L} \int_0^L 100 \sin \frac{n\pi x}{L} dx = \frac{200}{n\pi} (1 - (-1)^n) = \frac{400}{n\pi} \quad (n \text{ odd}), \quad 0 \quad (n \text{ even}).$$

$$u(x, t) = \frac{400}{\pi} \sum_{k \geq 0} \frac{1}{2k+1} \sin \frac{(2k+1)\pi x}{L} e^{-\alpha^2((2k+1)\pi/L)^2 t} \xrightarrow{t \rightarrow \infty} 0.$$

Solution 10. With initial velocity $g \equiv 0$, d'Alembert's formula reduces to

$$u(x, t) = \frac{1}{2} [f(x - ct) + f(x + ct)].$$

The initial shape splits into two half-amplitude copies, one traveling right at speed c and one left at speed c ; their sum reconstructs f at $t = 0$ and separates thereafter.

Unit Key Takeaway

One idea organizes the whole unit: **decompose into frequencies, then solve frequency by frequency**. Fourier series rebuild a function from orthogonal sines and cosines, with symmetry choosing the cosine or sine form and half-range extensions adapting it to a boundary. For PDEs, **separation of variables** turns the equation into the Unit 17 **eigenvalue BVP** plus a time ODE, superposition assembles the modes, and the **initial condition** is fitted by a Fourier series. The **Laplacian** is the shared spatial operator behind the heat, wave, and Laplace equations — diffusion that smooths, vibration that propagates, and the harmonic steady state.